# Harshit Bokadia

Applied Scientist — World Models · Reinforcement Learning · Probabilistic Modeling · Computational Neuroscience · Cognitive Science

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| scholar.google.ca/citations?user=QxgwOOcAAAAJ&hl=en&oi=ao | github.com/hbk008


## Summary


- Applied AI Scientist with 7+ years of experience designing, implementing, and deploying cutting-edge AI/ML systems in healthcare and regulatory environments.
- Strong foundation in both foundational ML research and applied problem-solving: independently driven end-to-end AI pipelines (problem abstraction → data engineering → model architecture design → rigorous evaluation → deployment)
- Currently at SUTD (Singapore, MoE-funded project): real-time estimation and scaffolding of learners' internal world models using Bayesian inverse RL over Markov Decision Processes, inferred from high-dimensional behavioral data.
- *Probabilistic modeling and interpretability foundations:* Novel perceptual and semantic interpretability metrics for saliency-based XAI; vector-symbolic architectures for structured knowledge encoding in transformer models.
- *Neuroscience and cognitive science foundations:* EEG-based brain-machine interface research; network-theoretic neural connectivity analysis; Neural ODE modeling of brain dynamics; Bayesian Teaching for human-AI decision-making.


**Research interests:** Model-based RL · World model estimation · Bayesian inference · Generative models · Vector-symbolic architectures · Representation learning · Neural dynamics · Interpretability

**Stack:** Python: PyTorch/ Numpy/Pandas/ Scikit-learn · Mujoco · MATLAB

**AI/ML publications:**

- **Bokadia, H.**, Yang, S. C. H., Li, Z., Folke, T., & Shafto, P. (2022). Evaluating perceptual and semantic interpretability of saliency methods: A case study of melanoma. *Applied AI Letters, 3*(3), e77. https://doi.org/10.1002/ail2.77

  Hu, B. X., Yu, T., Tuinstra, T., Rezai, R., **Bokadia, H.**, DiMaio, R., & Tripp, B. P. (2024). Encoding medical ontologies with holographic reduced representations for transformers. *ACM KDD Workshop on Knowledge-Infused Learning.* https://openreview.net/forum?id=LN4zA2D8vd

**Bio-signals/ Computational Neuroscience publications:**

- **Bokadia, H.**, Rai, R., & Torres, E. B. (2020). Digitized ADOS: Social interactions beyond the limits of the naked eye. *Journal of Personalized Medicine, 10*(4), 159. https://doi.org/10.3390/jpm10040159
- **Bokadia, H.**, Cole, J., & Torres, E. (2020). Neural connectivity evolution during adaptive learning with and without proprioception. *Proceedings of the 7th International Conference on Movement and Computing,* 1–4. https://doi.org/10.1145/3401956.3404232
- **All publications**: https://scholar.google.ca/citations?user=QxgwOOcAAAAJ&hl=en&oi=ao

## Education

**University of Texas, Austin** — **2024 - 2027**

**Master of Science- Artificial Intelligence (remote/ part-time)**

**Courses:** Ethics in AI, AI in Healthcare, Deep Learning, Advanced Deep Learning.

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**Rutgers University- New Brunswick, NJ, USA** — **2016 - 2018**

**Master of Science- Industrial & Systems Engineering**

**Thesis:** *Deep learning based virtual metrology for semiconductor manufacturing processes*

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**Rajasthan Technical University, India** — **2008 - 2012**

**Bachelor of Technology- Mechanical Engineering**

## Work Experience

**Singapore University of Technology and Design** — **March 2026 - Present**

**Research Associate (Full-time)**

- Built an **end-to-end embodied intelligence pipeline**: **MuJoCo humanoid simulation coupled with Bayesian inverse RL** to recover reward structure and transition dynamics from kinematic trajectories, grounding world model learning in physically plausible motor behaviour.

- Designed a **real-time world model inference loop over agent state-action spaces**; inferred latent reward functions and dynamics models to drive closed-loop, model-based RL scaffolding toward novel behavioural regimes.
- Processed 400+ **motion capture** sequences to build population-level behavioural priors; validated RL-inferred world model representations against real human movement data across participant cohorts.

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**Waypoint Centre for Mental Health Care, ON, Canada** — **May 2025 - Sept. 2025**

**Data Scientist (Full-time)**

- Led **AI strategy and governance for enterprise-wide adoption of AI solutions** to improve patient care and clinical decision-making.
- **Designed and deployed scalable AI solutions in production** using Azure ML, integrating predictive models with EHR systems for actionable insights while meeting regulatory standards.
- Conducted requirements gathering and stakeholder management across clinical and technical teams to translate healthcare needs into algorithmic solutions.

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**Holland Bloorview Kids Rehabilitation Hospital, Toronto, Canada** — **Jan 2023 - May 2025**

**Research Engineer (Full-time)**

- Developed statistical analysis and machine learning methodologies to support clinical research projects enabling data-driven insights for improved patient care.
- Designed and implemented an **AI-based medication recommendation system to augment clinicians' decision-making processes** improving treatment precision and outcomes (https://www.medrxiv.org/content/10.1101/2025.03.12.25323683v1).
- Led database development and management on REDCap for the Province of Ontario Neurodevelopmental (POND) network ensuring secure and scalable data pipelines for multi-site clinical trials.
- Utilized PostgreSQL to query large-scale datasets and provide actionable data exports across the POND network enhancing research efficiency and collaboration.

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**Mathworks x INCF, USA** — **July 2023 - Sept. 2023**

**Open source community toolbox internship (part-time/ remote)**

- Modeled large-scale neuronal data recordings to simulate brain network dynamics using MATLAB.
- Analyzed 2-photon imaging data to extract biologically meaningful brain activity patterns.
- Enhanced open-source MATLAB toolboxes for neuroscience research, ensuring scalability and usability.

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**Medical AI group, University of Waterloo, Ontario, Canada** — **May 2022 - Dec 2022**

**Research Assistant- II (Full-time) in the Department of Systems Design Engineering**

- Led a project on **analyzing FHIR data using deep learning and generative AI** in collaboration with a health-tech start-up.
- Developed a SNOMED-CT-based vector-symbolic language model **combining symbolic computation and deep learning** for healthcare applications.
- Worked with state-of-the-art transformer models to analyze electronic health records (MIMIC-IV dataset) improving clinical decision-making and healthcare outcomes.

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**Montreal Institute for Learning Algorithms (MILA), Montreal, Canada** — **July 2021 - Nov. 2021**

**Intern - NeuroAI (remote| volunteer work)**

- Conducted **large-scale analysis of neural dynamics using generative modeling and deep learning techniques.**
- Analyzed EEG data from healthy brain network (HBN) datasets with Python (MNE) uncovering patterns in neural activity.
- Applied Neural ODEs and dynamical systems modeling to simulate brain dynamics and predict complex neural behaviors.

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**Cognitive and Data Science Lab, Rutgers University-Newark, NJ, USA** — **Dec. 2020 - July 2021**

**Coadjutant (25 hours/week) in the Department of Mathematics and Computer Science**

- Developed PyTorch implementation of Probabilistic Linear Discriminant Analysis (PLDA) for medical imaging tasks.
- Applied **Bayesian Machine Teaching to combine cognitive science and machine learning enhancing human decision-making with AI systems.**
- Generated explanations for a VGG-16 model with attention modules for melanoma prediction on the ISIC dataset.
- Designed new metrics to evaluate interpretability of saliency methods improving trust and usability in AI-driven medical imaging.

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**Rutgers University- New Brunswick, NJ, USA** — **Sep. 2018 - Nov. 2020**

**Coadjutant (Full-time) at Rutgers Centre for Cognitive Science (Department of Psychology)**

- Conducted Brain-Machine Interface research, developing **neural correlates of disembodied cognition** by controlling cursor movement with EEG data under varying kinesthetic feedback conditions.
- Designed **digital biomarkers for Autism diagnosis** by analyzing wearable biosensor data during ADOS tests using network theory for connectivity analysis.

- Applied signal processing and statistical analysis to EEG and wearable biosensor data with MATLAB and EEGLAB uncovering stochastic patterns in socio-motor dyads.

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**UltraTech, Aditya Birla Group, India** **Sep. 2012 - Apr. 2014**

**Project Engineer (Full-time)**

- Analyzed data from **IoT sensors for Industry 4.0 projects optimizing process monitoring** and operational efficiency.
- Utilized control charts, statistical reliability analysis, and fault diagnosis to improve manufacturing workflows and reduce downtime.
- Conducted supply chain analytics to enhance resource utilization and production planning.
- Simulated production systems using Arena and Excel-VBA, identifying bottlenecks and evaluating process improvements.

## Community/ Volunteer work

- **Reviewer for IEEE Transactions on Industrial Informatics journal.**